DE–7390 **11**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization–Plant Biotechnology) DEGREE EXAMINATION, DECEMBER 2013.

PLANT DIVERSITY

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Give an account of the economic importance of cyanophyceae. (8)

   (b) Write a detailed account on the thallus organization in algae. (7)

   (c) Characterize the oogamous type of sexual reproduction. (5)

2. (a) What are the various types of thalli found in Fungi? (7)

   (b) Write about the spore dispersal mechanisms in Fungi. (7)

   (c) Brief on the characteristics of phycomycetes. (6)

3. (a) Give a short account on distribution of lichens. (6)

   (b) Explain the typical life cycle of a Bryophytes. (8)

   (c) Describe the structure of sporophyte in *Marchantia*. (6)

4. (a) Elaborate the stellar evolution in Pteridophytes. (8)

   (b) Explain the morphology of a lycopsida member. (6)

   (c) Give a general account on Fossil Pteridophytes. (6)

5. (a) What are the general characters of Gymnosperms? (8)

   (b) Brief on the classification of Gymnosperms. (8)

   (c) Compare the habit of Coniferales and Gnetales. (4)

6. (a) How the microalgae can be cultured? (6)

   (b) Give a brief account on reproduction in Fungi. (7)

   (c) Give a few economic importance of Lichens. (7)

7. (a) Compare the structure of thallus in *Anthoceros* and *Funaria*. (8)

   (b) Characterize the group Gnetales. (7)

   (c) Explain the process of fossilization. (5)

8. (a) Write about the life cycle pattern of a Cyanophyceae member. (6)

   (b) Outline the classification of Gymnosperms. (8)

   (c) Describe the fossil plant *Rhynia*. (6)

DE–7391 **12**

DISTANCE EDUCATION

M.Sc. DEGREE EXAMINATION, DECEMBER 2013.

Botany (Specialization Plant Biotechnology)

PLANT TAXONOMY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Explain in detail about the system of binomial nomenclature. (10)

   (b) Mention the kinds and particulars of species concept. (5)

   (c) Write the scope and application in the field of Taxonomy. (5)

2. (a) Give an account on Bentham and Hooker plant classification. (10)

   (b) Comment on phylogenetic relationship. (10)

3. (a) Describe biosystematics. (5)

   (b) Write the salient feature of plant geography and floreisties. (8)

   (c) Write short notes on chemotaxonomy. (7)

4. (a) What are the procedure should follow by publication of names? (6)

   (b) Comment on draft Biocode (8)

   (c) Why need for scientific name? Explain briefly. (6)

5. (a) Write an essay on diagnostic features of family *Ranunculaceae*. (10)

   (b) Write salient feature of *Boraginaceae* family (7)

   (c) Comment on ICBN. (3)

6. (a) Explain the theories of biological classification (10)

   (b) Write salient feature of family *Orchidaceae*. (10)

7. (a) Explain modern inter disciplinary approaches to taxonomy. (10)

   (b) What is the historical background of classification? (6)

   (c) Comment on 'Rejection of names'. (4)

8. (a) Write an essay on structural and biological systematics. (10)

   (b) Define ecotypes. (5)

   (c) Name few medicinal plants and uses from *Rhamnaceae* family. (5)

DE–7392 **13**

DISTANCE EDUCATION

M.Sc. Botany (Specialization in Plant Biotechnology) DEGREE EXAMINATION, DECEMBER 2013.

ANATOMY AND EMBRYOLOGY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carry 20 marks.

(5 × 20 = 100)

1. (a) Give an account on electron microscopic structure of cell wall. (8)

   (b) Write short note on specialization of phloem. (7)

   (c) Describe the mode of activity of vascular cambium. (5)

2. (a) What are the cambial variants? (6)

   (b) Write about the vasculature of leaf. (7)

   (c) Differentiate the vasculature in stem and root. (7)

3. (a) Write a brief account on identification of wood. (7)

   (b) Give an outline classification of woods. (8)

   (c) What are the uses of woods? (5)

4. (a) Write notes on development of anther. (8)

   (b) Describe various methods of vegetative reproduction. (7)

   (c) Write short note on nutrition of embryo sac. (5)

5. (a) Explain about apomixis. (7)

(b) Explain about apogamy. (7)

(c) Explain about poly embryony. (6)

6. (a) What are the molecular aspects of wood differentiation. (8)

(b) Elaborate on the pollen morphology. (7)

(c) Explain any two types of endosperms. (5)

7. (a) Explain about the phenomenon self-incompatibility. (7)

(b) Describe the root-stem transition. (7)

(c) Write a short note on apospory. (6)

8. (a) Describe the exploitation of polyembryony in higher plant improvement. (10)

(b) List down the important commercial woods of south India. (5)

(c) Explain about the root knots. (5)

———————

DE–7393 **14**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization Plant Bio-technology) DEGREE EXAMINATION, DECEMBER 2013.

PLANT TISSUE CULTURE

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

1. (a) What is media? Write about the composition of MS medium. (7)

   (b) What are the explants? How do you surface sterilise? (7)

   (c) Write short notes on Suspension culture. (6)

2. (a) Describe the Micropropagation. (8)

   (b) Give short notes on virus elimination in explants. (5)

   (c) Write short notes on : Growth regulators. (7)

3. (a) Write an account somatic embryogenesis. (8)

   (b) Describe the procedure of artificial seeds. (7)

   (c) Why should do plant tissue culture? Write about its uses. (5)

4. (a) Explain pollen culture techniques. (6)

   (b) Describe the various methods for callus regeneration. (6)

   (c) Write details about the different types of organ culture. (8)

5. (a) Describe the methods of protoplast isolation. (6)

(b) Explain secondary plant metabolites. (5)

(c) Write short notes :

(i) Fusagens (3)

(ii) Somaclonal variation (3)

(iii) Acclimatization. (3)

6. (a) Write about the mechanism and application of somoclonal variations. (10)

(b) Draw neat diagram of plant cell structure and write its functions. (5)

(c) Describe the status of plant tissue culture in India. (5)

7. (a) Short notes about application of somatic hybridisation. (10)

(b) What are the factors affecting production of somoclonal variants? (4)

(c) Explain embryo rescue? Write its applications. (6)

8. (a) Describe the methods of selection of hybrid protoplast. (7)

(b) Write about the types and application of artificial seeds. (7)

(c) Short notes on : Root culture. (6)

DE–7394 **15**

DISTANCE EDUCATION

M.Sc. (Botany- Specialization Plant Biotechnology) DEGREE EXAMINATION, DECEMBER 2013.

PLANT MOLECULAR BIOLOGY

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Draw the structure and describe the function of Chloroplast. (8)

   (b) How Chromatin organized in plant? (5)

   (c) Discuss about Nucleus encodes gene for Chloroplast proteins. (7)

2. (a) Give account on Mitochondria genome. (8)

   (b) Discuss about plant transposons with an example (8)

   (c) Give function of seed storage proteins (4)

3. Write short notes on

   (a) Agrobacterium and Crown Gal tumors. (6)

   (b) Binary vector. (6)

   (c) Give Mechanism of T DNA transfer. (8)

4. (a) Write short notes on (10)

      (i) Silicon Carbide W+11 SKER.

      (ii) Ultrasonication.

   (b) Give account on plant Stress response to Drought, Dehydration and UV. (10)

5. (a) Give account on Transgenic plant desired products for commercial application. (10)

   (b) Discuss about Bioremediation through plant. (5)

   (c) Explain Mapping. (5)

6. (a) Give details account on plant genome organization. (10)

   (b) Discuss about role of enzymes and protein in protein Targeting. (10)

7. (a) Write about the types seed storage proteins and give their classification. (15)

   (b) Explain about IBF and its function. (5)

8. (a) Describe the Regulation on Gene expression in plant development. (5)

   (b) Discuss about Plant Cloning vectors. (10)

   (c) Write notes on Agro Infection. (5)

DE–7395 **21**

DISTANCE EDUCATION

M.Sc. (Botany – Specialization Plant Biotechnology) DEGREE EXAMINATION, DECEMBER 2013.

PLANT PHYSIOLOGY AND BIOCHEMISTRY

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) What is active absorption and Passive absorption? Narrate the difference between them. (7)
   (b) What are the factors affecting the rate of transpiration? (7)
   (c) What is meant by transpirational pull? (6)

2. (a) Give an account of cohesion. (7)
   (b) Describe the mechanism of guttation. (6)
   (c) Explain the stomatal apparatus and its working (7)

3. (a) Enumerate and explain the types of C4 carbon cycles. (6)
   (b) Describe the events seen in biological nitrogen fixation. (7)
   (c) Narrate the processes taking place in the Hatch and Slack Cycle. (7)

4. (a) What are the factors affecting nutrient uptake in plants? (7)
   (b) Explain about the cyanide resistance respiration. (6)
   (c) Describe the mechanism of nitrate and ammonia assimilation. (7)

5. (a) Give an account of the significance of lipids. (7)

(b) Write short notes on the structure of nucleic acids. (7)

(c) Describe allosteric mechanism of enzymes. (6)

6. (a) Give an account of the classification of amino acids. (7)

(b) Write short notes on the primary and the secondary structure of proteins. (7)

(c) Define peptide bonds and enatiomers. (6)

7. (a) What is the significance of Michaelis Menten equation? (7)

(b) Explain the term isoenzyme, giving a suitable example. (7)

(c) Comment on enzymatic catalysis. (6)

8. (a) What is the fate of the byproducts of amino acid degradation? (7)

(b) How are amino acids classified? (7)

(c) Explain the processes of fatty acid oxidation. (6)

DE–7396 **22**

DISTANCE EDUCATION

M.Sc.(Botany) (Specialization–Plant Biotechnology) DEGREE EXAMINATION, DECEMBER 2013.

CELL BIOLOGY, GENETICS AND PLANT BREEDING

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry 20 marks.

$(5 \times 20 = 100)$

1. (a) Describe the structure and function the Golgi complex. (7)

   (b) Enumerate the functions of the nucleus. (6)

   (c) Describe the organization of a prokaryotic cell. (7)

2. (a) What are the postulates of the fluid mosaic model? (7)

   (b) What are membrane proteins and describe its types and functions? (6)

   (c) Describe the events taking place during prophase I of meiosis. (7)

3. (a) What is polyploidy? Describe the various types. (7)

   (b) List the applications of Hardy-Weinberg's law in calculating gene frequencies in a population. (6)

   (c) Enumerate the different types of mutagenic agents and add a note on each of the types. (7)

4. (a) Define Mendel's law of segregation and give a suitable example to explain the law. (7)

   (b) At what stage of meiosis does crossing over occur? Describe one of the classical experiments in which it was detected cytologically. (7)

   (c) With suitable example explain extra nuclear inheritance by cellular organelles. (6)

5. (a) What is genetic variability? Elaborate on its role in plant breeding. (7)

   (b) Describe a method by which a plant is bred for stress tolerance. (6)

   (c) Write a note on any one method employed for the breeding of self pollinated plants. (7)

6. (a) What is a point mutation? Describe the types of point mutations. (7)

   (b) Write short notes on gene pool and gene frequency. (6)

   (c) What are the methods by which polyploidy can be induced in plants? (7)

7. (a) What are Morgan's views on linkage? (8)

   (b) Write a short note on the process of synapsis, which occurs during meiocis. (6)

   (c) Explain test cross. (6)

8. (a) Comment on protein sorting in mitochondria. (7)

   (b) List out the functions of the plasma membrane. (7)

   (c) Describe the structure and functions of lysosomes.(6)

DE–7397 **23**

DISTANCE EDUCATION

M.Sc. (Botany- Specialization Plant Biotechnology)DEGREE EXAMINATION, DECEMBER 2013.

BIOTECHNIQUES IN BOTANY

(2008 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Principle and application of PAGE. (8)

   (b) What are the applications of 2D electrophoresis. (7)

   (c) Write the principle and application of agarose gel electrophoresis. (5)

2. (a) Write the principle of centrifugation. (7)

   (b) Write the various types of electron microscopes. (8)

   (c) Write the principle of HPTLC. (5)

3. (a) Comment on Scintillation counting. (6)

   (b) Write notes on radioactive isotope. (7)

   (c) Write notes on autoradiography. (7)

4. (a) Give a critical comment on GCMS. (7)

   (b) Discuss about the various types of PCR and their applications. (7)

   (c) Explain about AFLP. (6)

5. (a) How will you apply the various bioinformatic tools in botany. (6)

(b) Write brief notes on DNA fingerprinting (7)

(c) Write about the different types of blotting techniques and their principle. (7)

6. (a) State the relationship between genomics and proteomics. (7)

(b) Explain about TEM. (7)

(c) How will you track gene expression in plant cells?(6)

7. (a) Write short notes on microarrays. (7)

(b) Write detailed notes on isoelectric focusing. (6)

(c) Comment on phase contrast microscopy. (7)

8. (a) Write the principle of hybridization technique. (6)

(b) Write notes on MALDI- TOF. (6)

(c) How will you examine the plant tissue under electron microscope? What are the procedures involved in the sample preparation. (8)

———————

DE–7398 **24**

DISTANCE EDUCATION

M.Sc. (Botany) DEGREE EXAMINATION, DECEMBER 2013.

Specialization : Plant Biotechnology

BIODIVERSITY CONSERVATION AND IPR

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$)

1. (a) Write brief notes on ecological and genetic diversity. (4)

   (b) Write critical notes on hot spots. (6)

   (c) Describe the different causes of loss of biodiversity. In what ways it can be checked? (10)

2. (a) What are the common features of threatened plants? (6)

   (b) Distinguish between endangered plants and vulnerable plants. (6)

   (c) Write a brief note on red data book. (8)

3. (a) What are field gene banks? Mention their operational procedures. (6)

   (b) Write a brief account on biosphere reserves and their role in conservation of biodiversity. (6)

   (c) Enumerate the importance of tissue culture technology in plant conservation. (8)

4. (a) Explain the main features of GATT. (5)

   (b) Discuss the advantages and disadvantages of patent system. (8)

   (c) Give an account of plant breeders right. (7)

5. (a) Write a brief note on Agro biodiversity. (5)

   (b) Discuss in detail about values and uses of biodiversity. (5)

   (c) Describe the various phytogeographic regions of India. (10)

6. (a) Write a critical notes on mangrove forests. (6)

   (b) Describe the vegetation of Deccan. (7)

   (c) Give an account of grassland vegetation. (7)

7. (a) Write a brief note on WTO. (5)

   (b) Comment on copy right. (7)

   (c) Discuss the general conditions for working in transgenic plants. (8)

8. (a) Describe the patent case studies of basmati rice. (5)

   (b) Enumerate the conditions for patenting GM food crops. (5)

   (c) Point out the good laboratory practices for a biological lab. (10)

DE–7399 **25**

DISTANCE EDUCATION

M.Sc. (Botany) (Specialization–Plant Bio-technology) DEGREE EXAMINATION, DECEMBER 2013.

PLANT GENETIC ENGINEERING

(2008 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Give their important historical perspective in genetic engineering. (8)

   (b) Draw the diagrammatic representation of gene cloning? (6)

   (c) Write short notes on alkaline phosphatase, linker and adaptors. (2 + 2 + 2)

2. (a) Give their protocol to antibiotic sensitivity assay. (6)

   (b) Write short notes on selectable marker gene. (6)

   (c) Explain about 35S promoter and draw the diagram. (8)

3. (a) Write notes on particle bombardment. (6)

   (b) Give account on electroporation. (6)

   (c) Discuss about golden rice and its importance. (8)

4. (a) Write detailed about herbicide resistance transgenic. (6)

   (b) Describe mechanism of transgenic BT cotton. (6)

   (c) What are the gene involved in delayed fruit ripening and explain? (8)

5. (a) Discuss about transplastomic plant. (8)

   (b) What is molecular marker and explain STS and microsatellite? (8)

   (c) What are advantages of chloroplast engineering? (4)

6. (a) Discuss the merit and Demerit plant genetic engineering. (8)

   (b) Explain shuttle vector. (6)

   (c) Explain about tissue specific promoters. (6)

7. (a) Draw the diagram and explain YAC. (6)

   (b) How will you selection of recombinants through blue and white colony method and explain? (8)

   (c) Explain southern blotting techniques. (6)

8. (a) Describe the methods of abiotic stress tolerance to plant? (8)

   (b) Discuss about RNAi interferon's. (6)

   (c) Write about AFLP. (6)